BRIAN CAO

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Education

Stony Brook University **Expected May 2027**

Majors (B.S.): Computer Science Honors GPA: 3.86; Dean's List

Coursework: Algorithms, Deep Learning, Software Eng., Systems, Linear Algebra, Graph Theory, Probability, Security

Experience

Stony Brook University Computer Science Department – Operations Stony Brook, NY

Software Developer and IT Support; (Python, Zabbix, Linux, Network Monitoring, Automation, Bash) **Sep 2025–Present**

- Automated data processing pipelines in Python for graduate application workflows, reducing manual review time by 60% and improving system maintainability through modular architecture
- Configured and troubleshot 10+ Power Distribution Units (PDUs) for network device monitoring, resolving authentication, firmware, and polling issues to improve infrastructure reliability
- Integrated 20+ campus devices into the Zabbix monitoring system for network monitoring, reducing issue detection time
- **Impact**: Improved reliability and observability of campus infrastructure by automating workflows and enabling proactive monitoring across 20+ devices and critical systems

Pulp Corporation Brooklyn, NY

Software Engineer Intern; (Next.js, TypeScript, Neo4J, AWS) **Jan 2025–Aug 2025**

- Engineered a full-stack job-matching platform using Next.js, TypeScript, and Neo4j, leveraging graph-based data modeling for backend systems to automate matching for 100+ users and reduce manual review time by 75%
- Implemented data ingestion pipelines for CSV parsing, validation, and transformation with REST APIs, improving data processing efficiency and increasing onboarding speed by 300%
- Built and standardized reusable UI components using design tokens and Storybook, improving frontend system consistency and development efficiency by 200%
- **Impact**: Streamlined job matching and onboarding workflows for 100+ users, significantly reducing manual review effort and improving end-to-end platform efficiency

Stony Brook University Computer Science Department - Data Stony Brook, NY

Web Programmer; (Drupal, PHP, Twig, HTML/CSS, JavaScript) **Jan 2025–Aug 2025**

- Migrated 5+ departmental websites to Drupal 10 CMS, improving performance and accessibility scores by 40%
- Built reusable Twig components in Drupal to improve component reusability, reducing user issues by 70%
- **Impact**: Improved accessibility and usability for 500+ students and faculty, reducing friction in accessing course and departmental resources

Stony Brook University Campus Residences Stony Brook, NY

Resident Assistant; (Event Planning, Management, Operations) **Aug 2024–Present**

- Spearheaded 5 campus-wide events with over 600 attendees collaborating with 10+ RAs, 650% more than average
- Supervising 40 residents consisted of first year, international and transfer students to foster community building
- Handled and mediated 5 roommate conflicts to ensure the inclusivity inside dorm building
- **Impact**: 99% retention rate of residents returning next year and gained $1,000 extra funding for future events

Projects

Financial News Sentiment Analysis with RNNs (Python, PyTorch, NLP) **Dec 2025–Present**

- Developed and trained RNN/GRU models on the FinancialPhraseBank dataset to classify financial headline sentiment (positive, negative, neutral)
- Built a preprocessing pipeline including tokenization, padding, and word embeddings to convert raw text into model sequences
- Evaluated model performance using accuracy, precision, recall, and F1-score; optimized hyperparameters to improve classification results
- Analyzed sentiment trends across financial headlines to extract insights relevant to market perception and decision-making

Playlister (React, Node.js, Express, MongoDB) **Aug 2025–Dec 2025**

- Implemented client-side state management to handle complex UI behavior, including undo/redo functionality, edit modals, and responsive components
- Designed RESTful API endpoints using Express and structured MongoDB schemas to support playlist CRUD operations and embedded song data
- Built a full-stack playlist management application with React and Node.js, enabling dynamic user interactions with real-time state synchronization

Technical Skills

Languages: Python, SQL, PostgreSQL, C/C++, Java, JavaScript, TypeScript, R, OCaml, HTML, CSS

Libraries/Tools: AWS, Hugging Face, Transformer Library, PyTorch, Jupyter, Numpy, Pandas, Bash, Git, Flask, Bootstrap, .NET, LaTeX